

HALIUT, China

WMO: 533360

Lat: 41.5661N

Lon: 108.5078E

Elev: 1290

StdP: 86.76

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-20.5	-30.0	0.3	-17.8	-28.2	0.3	-15.9	10.4	-13.2	8.9	-10.8	1.7	40	0.542

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	31.6	16.5	30.3	16.2	29.0	15.7	20.0	26.7	18.9	25.8	18.0	24.9	4.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.8	15.0	23.6	16.6	13.9	22.2	15.6	12.9	21.5	63.7	26.4	59.6	25.7	56.3	24.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.3	7.2	DB	-25.2	34.0	2.6	1.7	-27.0	35.3	-28.5	36.3	-30.0	37.2	-31.9	38.5	(4)
			WB	-25.4	21.2	2.4	1.1	-27.1	22.0	-28.5	22.7	-29.8	23.3	-31.5	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	6.6	-12.2	-7.6	0.3	8.8	15.2	20.6	23.2	21.1	15.0	6.9	-2.6	-10.2	(6)
(7)		DBStd	12.91	4.06	4.96	5.29	5.05	4.17	3.09	2.50	2.84	3.78	4.42	4.72	4.12	(7)
(8)		HDD10.0	2693	688	492	302	80	8	0	0	6	113	378	627		(8)
(9)		HDD18.3	4629	946	725	559	287	114	14	1	8	110	353	628	885	(9)
(10)		CDD10.0	1457	0	0	2	43	168	318	408	343	157	18	0	0	(10)
(11)		CDD18.3	353	0	0	0	1	16	82	151	92	11	0	0	0	(11)
(12)		CDH23.3	3218	0	0	0	22	208	791	1387	716	93	1	0	0	(12)
(13)		CDH26.7	943	0	0	0	2	40	228	481	178	13	0	0	0	(13)
(14)	Wind	WSAvg	2.8	2.3	2.6	3.2	3.7	3.2	2.8	2.5	2.6	2.6	2.5	2.3	(14)	
(15)	Precipitation	PrecAvg	200	2	1	3	7	11	24	60	56	23	11	2	1	(15)
(16)		PrecMax	433	6	5	16	37	56	113	208	158	63	32	18	5	(16)
(17)		PrecMin	87	0	0	0	0	0	1	3	9	0	0	0	0	(17)
(18)		PrecStd	76	2	2	4	9	14	22	48	33	18	9	4	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	1.6	7.6	18.0	26.3	30.0	32.6	33.9	31.9	28.4	21.7	11.6	2.1	(19)	
(20)		MCWB	-3.1	0.4	6.2	10.6	12.8	15.2	18.9	17.2	14.6	10.6	4.5	-2.5	(20)	
(21)		DB	-1.3	4.9	14.2	22.9	27.6	30.8	32.1	30.1	25.7	19.0	9.1	-0.3	(21)	
(22)		MCWB	-5.1	-1.1	4.4	9.3	12.1	14.8	17.2	17.3	13.9	9.1	2.5	-4.0	(22)	
(23)		DB	-3.4	2.7	11.6	20.2	25.7	29.3	30.8	28.7	24.0	16.8	7.1	-1.9	(23)	
(24)		MCWB	-6.6	-2.2	2.8	7.9	11.2	14.4	17.1	16.7	12.8	8.1	1.3	-5.0	(24)	
(25)		DB	-5.2	0.7	9.1	17.9	23.7	27.7	29.5	27.2	22.2	14.7	5.0	-3.5	(25)	
(26)		MCWB	-7.9	-3.4	1.5	6.8	10.2	13.9	16.6	16.2	12.3	7.0	0.3	-6.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	-3.0	1.6	7.0	12.3	14.9	18.4	21.9	21.6	17.8	12.3	5.3	-1.9	(27)	
(28)		MCDWB	1.1	5.3	16.1	21.4	23.6	25.3	30.5	27.9	22.3	18.1	10.4	1.1	(28)	
(29)		WB	-5.0	-0.3	4.8	10.4	13.4	17.2	20.3	20.1	16.0	10.5	3.3	-3.6	(29)	
(30)		MCDWB	-1.5	3.8	13.1	19.3	23.5	24.7	28.2	25.8	21.9	16.9	7.9	-0.7	(30)	
(31)		WB	-6.5	-1.9	3.3	8.9	12.4	16.3	19.2	18.9	14.7	9.1	1.7	-4.9	(31)	
(32)		MCDWB	-3.5	2.6	10.7	18.8	22.3	24.5	26.6	25.2	20.8	15.0	6.7	-2.0	(32)	
(33)		WB	-7.9	-3.5	1.8	7.4	11.4	15.5	18.4	17.8	13.7	7.8	0.3	-6.2	(33)	
(34)		MCDWB	-5.3	0.7	8.6	17.2	21.4	24.2	25.5	24.5	20.1	13.4	4.7	-3.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.4	11.5	12.2	12.3	12.2	11.3	10.5	10.3	11.0	11.5	10.3	9.7	(35)	
(36)		5% DB	12.1	13.3	15.1	15.4	14.8	13.7	12.4	11.9	13.0	13.7	12.4	10.8	(36)	
(37)		MCWBR	9.2	9.1	8.8	7.7	6.5	4.7	4.2	4.4	5.5	6.9	7.6	8.0	(37)	
(38)		5% WB	11.9	12.6	13.9	13.2	12.2	10.9	9.9	9.3	10.4	11.8	11.7	10.6	(38)	
(39)		MCWBR	9.1	8.8	8.4	7.2	6.0	4.3	4.2	4.6	5.2	6.7	7.5	8.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.331	0.377	0.430	0.464	0.492	0.465	0.458	0.423	0.368	0.316	0.291	(40)	
(41)		taud	2.230	2.129	2.013	1.923	1.897	1.900	2.043	2.040	2.058	2.135	2.204	2.209	(41)	
(42)		Ebn at Noon	849	860	857	837	817	793	811	804	802	803	803	809	(42)	
(43)		Edh at Noon	93	122	155	183	193	193	165	161	147	120	95	88	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.39	3.48	4.94	6.16	6.98	6.75	6.46	5.96	4.94	3.98	2.70	2.13	(44)	
(45)		RadStd	0.24	0.24	0.31	0.31	0.35	0.42	0.35	0.22	0.24	0.16	0.23	0.24	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-1.23	N/A	N/A	N/A	N/A	N/A	+83	N/A	(46)
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.45	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page